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Even and odd subharmonic frequencies and chaos in Josephson junctions: Impact on parametric amplifiers?

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The Stewart-McCumber model of a Josephson junction has been shown to exhibit period-doubling bifurcation cascades, as described by the Feigenbaum bifurcation theory. Chaotic states, sometimes associated with the bifurcations, are also prevalent. The present paper deals with the questions of subharmonic generation and chaotic states in the aforementioned model, and in addition with the problem of the ubiquitous noise rise found in Josephson junction parametric amplifiers. The bifurcation is first discussed by drawing on analytical results on the Duffing equation which is an approximation to the complete ac-driven Stewart-McCumber model. The complete model is then solved on an analog computer. Thereafter it is shown that besides the even subharmonics predicted by the bifurcation theory, the natural subharmonic to expect at small dc currents is the odd. This may then have associated its own bifurcation tree. The role of spontaneous symmetry breaking will be discussed. This reconciles the earlier treatment of the 3-photon amplifier with the Feigenbaum scheme. Finally, analog calculations on a model of an externally pumped Josephson junction parametric amplifier will be discussed. The conclusion seems to be that chaotic noise cannot account for the noise rise.

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I. INTRODUCTION

The existence of period-doubling bifurcation cascades in the Stewart-McCumber¹ model of a Josephson junction has been pointed out by Huberman *et al.*² Since then a rather extensive study of the conditions for the cascading bifurcations in this model (see Fig. 1) has been performed by Kautz.³ It is the aim here first to provide a more detailed understanding of the connection between the earlier theories⁴ for parametric amplification on the supercurrent (and the microwave induced constant voltage steps) and the cascading bifurcation theory of Feigenbaum.⁵ Especially the role of spontaneous symmetry breaking will be treated in detail. In Ref. 4 it was shown that the ac-driven Stewart-McCumber model could be unstable towards generation of a half-harmonic. For this to happen conditions were $i_{dc} \neq 0$ and the pump frequency f_D approximately equal to twice the Josephson plasma frequency f_p . The condition $i_{dc} \neq 0$ came about by proving that the threshold pump power for $i_{dc} = 0$ was infinite. In Ref. 6 a symmetry argument was similarly employed to rule out the possibility of 3-photon interaction for $i_{dc} = 0$. Since in Ref. 2 it was found that bifurcation took place for $i_{dc} = 0$ and a condition was $f_D < f_p$, obviously some clarification is needed for surely the two phenomena are one and the same. Secondly, it will be demonstrated that the natural subharmonic to expect is really the odd subharmonic and not the even that comes about as an instability. Associat-

ed with this odd subharmonic may then again be a Feigenbaum bifurcation tree.

In Ref. 2 a very complicated pattern of deterministic chaos, sometimes associated with the bifurcation cascades, was also observed. This chaotic noise was pointed to in Ref. 2 as a likely source of the noise rise always encountered in externally pumped Josephson junction parametric amplifiers. However, it is of course not enough to show the existence of chaotic states. In order to test the idea a complete amplifier was modeled on an analog computer. Both the 3- and 4-photon amplifier cases were treated. Also the existence of the diverse subharmonics was investigated with the help of the analog computer. As in the cases of Refs. 2 and 3 the basic junction model will be the ac-driven Stewart-McCumber model. This is rather difficult to treat analytically and we shall therefore take recourse to the Duffing equation to give a qualitative understanding of what to expect and then use an analog computer (EMRI-ACS-122 D hybrid analog computer) to verify the conclusions on the complete model.

II. SUBHARMONIC GENERATION

The basic ingredients in the Stewart-McCumber model are shown in Fig. 1. In terms of dimensionless parameters the differential equation for this model reads

$$\beta_c \ddot{\phi} + \dot{\phi} + \sin \phi = i_{dc} + i_D \cos \Omega_D \tau, \quad (1)$$

where $\beta_c = 2eR^2I_cC/\hbar$, $\Omega_D = (\hbar/2eRI_c)\omega_D$, and $\tau = (2eRI_c/\hbar)t$. Currents and voltages are normalized by I_c and RI_c , respectively. In normalized units the Josephson plasma frequency $\omega_p = (2eI_c/\hbar C)^{1/2}$ becomes $\Omega_p = \beta_c^{-1/2}$. Since as stressed above the equation is rather unhandy we shall in the following turn to Duffings equation. This comes about by expanding $\sin \phi$ and retaining only the first two terms. Written in the usual notation⁷ we have

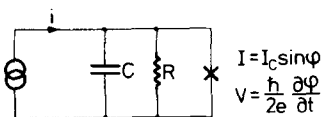


FIG. 1. Junction equivalent diagram. The current source can be both dc and ac. $I = I_{dc} + I_D \sin \omega_D t$.

$$\ddot{v} + 2\delta\dot{v} + c_1v + c_3v^3 = B_0 + B \cos \nu t, \quad (2)$$

where ν is an integer.

In what follows we shall assume that $2\delta \ll 1$, i.e., the junction is highly undamped. The behavior of this equation is well documented in the literature⁷ and it is therefore well suited for expounding the qualitative behavior to be expected from the Stewart-McCumber model. Following Huberman *et al.*² let us first assume $B_0 = 0$, and then guess as an approximate solution

$$\begin{aligned} v &= v_0 + \xi \\ &= z + w \cos \nu \tau + \xi. \end{aligned} \quad (3a)$$

Here v_0 is an appropriate steady-state solution before some (small) change in a parameter introduces a change ξ . z is a dc term. Under the aforementioned condition this trial function is a quite good approximation. The amplitude w is furthermore well approximated by

$$w = \frac{1}{1 - \nu^2} B. \quad (4)$$

If one now suspects the existence of a subharmonic Eq. (3a) is written explicitly as

$$v = z + w \cos \nu \tau + x \sin \tau + y \cos \tau. \quad (3b)$$

By reason of symmetry one would guess that $z = 0$ is a necessary condition. Using this, insertion of Eq. (3b) into Eq. (2) under the condition $\nu \neq 3$, yields (by collecting terms in $\sin \tau$ and $\cos \tau$)

$$\begin{aligned} c_3 \left[1 - \frac{3}{4}(x^2 + y^2) - \frac{3}{2}w^2 \right] x + 2\delta y &= 0, \\ c_3 \left[1 - \frac{3}{4}(x^2 + y^2) - \frac{3}{2}w^2 \right] y - 2\delta x &= 0. \end{aligned} \quad (5)$$

From this set of equations we find

$$2\delta(x^2 + y^2) = 0 \quad (6)$$

for the amplitude of the subharmonic oscillation. This implies that no subharmonic oscillation can occur for $B_0 = 0$ unless no damping is present in agreement with Refs. 4 and 6 and in seeming disagreement with Ref. 2. The solution to this discrepancy is broken symmetry. As noted in Ref. 7 it is really the choice of $z = 0$ that results in the banishment of subharmonics. Another known case where broken symmetry plays an important role in the behavior of the Josephson junction is $i_{dc} = 0$, $\phi = n\omega t$, i.e., the spontaneous jump to zero-crossing steps.⁸ This is what is called a traveling limit cycle in Ref. 2.

III. EVEN SUBHARMONICS

If we now concern ourselves with the case $\nu = 2$ and allow $z \neq 0$, the following terms are added to the left hand side of Eq. (5)

$$\begin{aligned} &+ c_3(3z^2x - 3zwx) \\ &+ c_3(3z^2y + 3zwy). \end{aligned} \quad (7)$$

This results in the condition [instead of Eq. (6)]

$$2\delta(x^2 + y^2) = 6c_3zwx. \quad (8)$$

Also one finds for the dc term the equation

$$c_1z + c_3 \left[z^3 + \frac{3z}{2}(x^2 + y^2) + \frac{3z}{2}w^2 - \frac{3}{4}w(x^2 - y^2) \right] = 0. \quad (9)$$

Thus the existence of a half-harmonic is possible if at the same time $z \neq 0$. Since this is really a bifurcation there is of course two values of z : $z_1 = -z_2$, giving symmetric solutions. In the language of the Josephson junction this means that

$$i_{dc} = \langle \sin \phi \rangle = 0 \quad (10)$$

does not necessarily imply

$$\langle \phi \rangle = 0 \quad (11)$$

which at first seems rather strange and explains why this solution was ruled out in Refs. 4 and 6. From Eq. (9) it is seen that $z \neq 0$ is actually a second order effect in ξ . The possibility of $z \neq 0$ was therefore ruled out by the linearization of the Stewart-McCumber equation performed in Ref. 4. If the second order term is retained it is in fact easy to see by guessing a trial function that includes a half-harmonic, the relaxation of the condition in Eq. (11). In agreement with Ref. 2 this is seen to imply rather high power levels (inherent in a second order effect) where the supercurrent is very much depressed and so the plasma frequency. It is therefore not surprising that the half-harmonic generation for $i_{dc} = B_0 = 0$ only seems to happen when $\omega_D < \omega_p$ (ω_p being the undisturbed plasma frequency). Knowing this one might easily, by playing with a pendulum, convince oneself about the possibility of the $z \neq 0$ solution (or in the Josephson language $\langle \phi \rangle \neq 0$ solution). From Eq. (9) one can easily see that for finite B_0 ($i_{dc} \neq 0$) one no longer has to take the second order term into account. The result is that the excitation of the half-harmonic can take place for much lower power values. This then explains why the frequency condition can be $\omega_D \sim 2\omega_p$. In this case the symmetry is already broken by the dc-current term.

Let us now briefly discuss the case of $\nu = 4$. For $\nu = 4$ and $z \neq 0$ we have to add only

$$\begin{aligned} &+ c_3 3z^2x \\ &+ c_3 3z^2y \end{aligned} \quad (12)$$

to the left hand side of Eq. (5). Thus we again find the condition Eq. (6) for the amplitude; i.e., only for $\delta = 0$ (no damping) can a $1/4$ subharmonic exist in the Duffing equation without a $1/2$ subharmonic being present. Inclusion of the $1/4$ subharmonic in the steady-state solution will of course through some rather tedious calculations show that the $1/4$ subharmonic is a possibility even for $\delta \neq 0$. More about this in Sec. IV.

The usual procedure for calculating the stability of a steady-state solution is to go back to Eq. (3a) and treat ξ as a small perturbation. Linearization of the equation in ξ makes it possible to transform the Duffing equation into a Mathieu equation which may be unstable towards excitation of the half-harmonic. With the transformation $\xi = e^{-\delta\tau}\eta$ this reads

$$\frac{d^2\eta}{d\tau^2} + (c_1 - \delta^2 + 3c_3v_0^2)\eta = 0. \quad (13)$$

If the half-harmonic then is stable one includes this in the steady-state solution v_0 . The linearization then allows the transformation into a Hill's type equation

$$\frac{d^2\eta}{d\tau^2} + \left[\theta_0 + 2 \sum_{n=1}^3 \theta_n \cos(2n\tau - \epsilon_n) \right] \eta = 0, \quad (14)$$

where the specific values of the constants θ_n and ϵ_n need not concern us.

This equation may be unstable towards a $1/4$ subharmonic. If this then is stable it can be incorporated in v_0 and so on *ad infinitum*. This is the origin of the cascading bifurcation of Feigenbaum.⁵ The Mathieu equation was already utilized for the McCumber case in Ref. 4 working on the 3-photon amplifier. We should like to remark here that if one sets $B_0 = z = 0$ in Eq. (13) then the driving term in v_0^2 has the frequency $2\omega_D$. The system is therefore unstable towards ω_D ! This is the basis for the 4-photon parametric amplifier.^{6,9} Of course no bifurcation is connected with this instability for $z = 0$.

Use of the Duffing equation can naturally only give an idea of what may be expected from the full equation. However, it is already known from Refs. 2 and 3 that the bifurcation actually may occur. The analog calculations shown in Fig. 2 (spectrum obtained on a 3582A H.P. spectrum analyzer) are

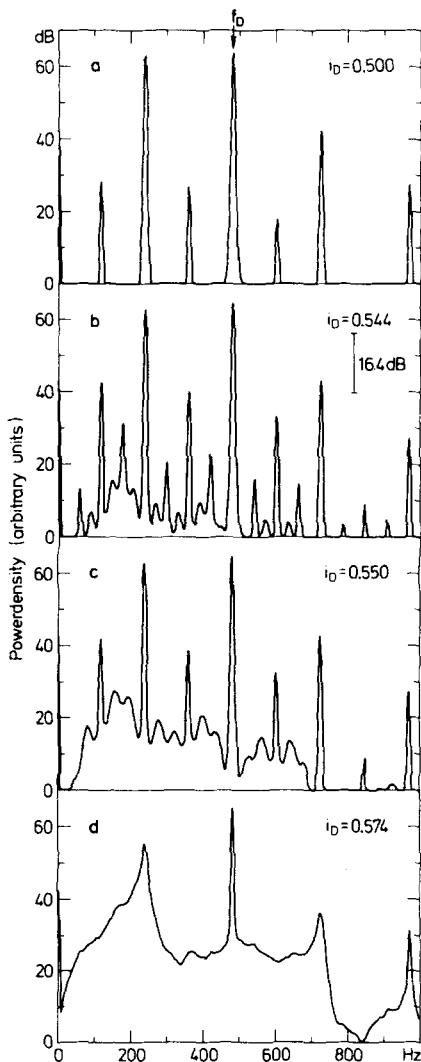


FIG. 2. Examples of subharmonic generation in a cascading bifurcation leading to chaos. "Junction parameters": $\beta_c = 25$, and $f_p = 318$ Hz. $i_{dc} = 0.5$, $f_D = 480$ Hz. The vertical bar in 2(b) shows the expected drop in successive power levels of subharmonics. A chaotic trace is shown in Fig. 2(d).

meant to illustrate two important features. First, that the cascading bifurcation may actually happen for $\omega_D \sim 2\omega_p(i_{dc} \neq 0)$ and therefore in a regime of interest for the 3-photon amplifier. Secondly, Fig. 2(c) shows an unexpected feature, namely what looks like a $1/12$ subharmonic. This may of course be attributed to the finite precision of the analog computer, but turning back to the hitherto neglected case of $\nu = 3$ in the treatment of the Duffing equation another explanation will be seen to prevail. This will be the subject of Sec. IV.

Concluding Sec. III we should like to remark on the comparison with the Feigenbaum theory. Two universal predictions are resulting from this. This first is that if a change in a parameter α brings about a cascade, then the critical values α_n for the 2^{-n} subharmonic to occur form a geometric series such that $\lim_{n \rightarrow \infty} \alpha_n - \alpha_{n-1} / \alpha_{n+1} - \alpha_n = \delta = 4.669...$ This has been shown to be fulfilled in some cases for the ac-driven Stewart-McCumber model both for $\alpha = i_D$ (Ref. 2) and $\alpha = i_{dc}$ (Ref. 3). However, the occurrence of the $1/12$ subharmonic in Fig. 2(c) shows that the picture may be much more complicated than hitherto conjectured, since the $1/12$ subharmonic actually breaks into the bifurcation cascade just after the $1/32$ subharmonic has appeared. The chaotic state still occurs but at a higher power level than expected.

The other prediction concerns the power levels of the subharmonic which should differ by approximately 16.4 dB for adjacent values of n . From Fig. 2(b) this also is seen to be approximately correct.

IV. ODD SUBHARMONICS

Returning to the discussion in Sec. II we shall now consider the case of $\nu = 3$. For simplicity we take $B_0 = 0$. Again we use a trial function of the form given in Eq. (3b). In this we set $z = 0$. As will be seen this does not have the disruptive effect as was the case for all other subharmonics. Substitution of the trial function into the Duffing equation leads to (again isolating terms in $\sin \tau$ and $\cos \tau$)

$$\begin{aligned} Ax + 2\delta y &= -\frac{3w}{2}c_3xy, \\ Ay - 2\delta x &= -\frac{3w}{4}c_3(x^2 + y^2), \end{aligned} \quad (15)$$

where

$$A = 1 - c_1 - \left[\frac{3}{4}(x^2 + y^2) + \frac{3}{2}w^2 \right] c_3. \quad (16)$$

By squaring and adding the two equations constituting Eq. (15) we obtain

$$A^2 + (2\delta)^2 = c_3^2 \left(\frac{3w}{4} \right)^2 (x^2 + y^2). \quad (17)$$

Therefore a $1/3$ subharmonic may occur *whether or not* $\nu \neq 0$. It is easily shown that this is also the case for $B_0 \neq 0$.

This is a direct consequence of the relation

$$\sin 3z = 3 \sin z - 4 \sin^3 z. \quad (18)$$

From this one can immediately see why a $1/2$ subharmonic can be a natural frequency to expect for $B_0 \neq 0$. By a suitable choice of k , $v = u + k$ transforms the Duffing equation with

$B_0 \neq 0$ into the form⁷ (corresponding to the expansion of $\sin \phi$ around $\phi_0 \neq 0$)

$$\ddot{u} + 2\delta\dot{u} + c_1u + c_2u^2 + c_3u^3 = B \cos vt. \quad (19)$$

The leading nonlinear term is thus in u^2 . No spontaneous symmetry breaking is therefore required. This is the basis of the optical parametric amplifier with the obvious implication that a similar bifurcation cascade may be possible here.

Since

$$\sin nz = \sum_{k=1}^{(n-1)/2} a_{2k-1} \sin(2k-1)z + a_n \sin^nz \quad (20)$$

including a v^n nonlinearity in the Duffing equation might lead one to suspect that $1/n$ subharmonics may also occur given suitable conditions. This can indeed be proven as above and a general statement is as follows. If the maximum nonlinear term is of order n , n odd, then subharmonics up to but not above n th order may occur! This has the following interesting consequence for the Josephson junction. Since the expansion for $\sin \phi$ is an infinite series in odd powers, all kinds of odd subharmonics may be expected to occur depending on the damping δ . Whether this conjecture is true must of course be investigated using the complete McCumber equation. Figure 3 shows different types of behavior calculated on the analog computer using the full equation, showing that odd subharmonics are indeed the natural subharmonics to expect for the Josephson junction at $i_{dc} = 0$. A comment on the absence of odd subharmonics in the calculation by Kautz³ is relevant here. The drive frequency chosen by Kautz was equal to the plasma frequency. We found that somewhat lower frequencies were needed to

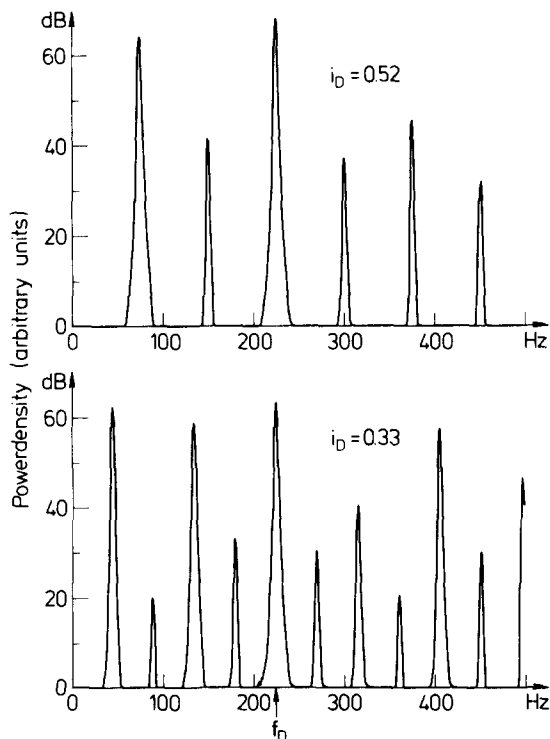


FIG. 3. Examples of generation of odd subharmonics. "Junction parameters" as Fig. 2. $i_{dc} = 0.4$. $f_D = 210$ Hz.

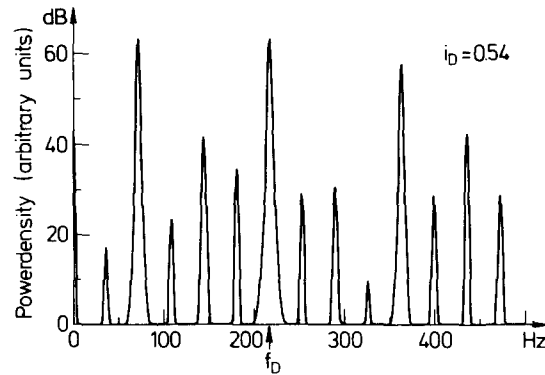


FIG. 4. Start of bifurcation from an odd subharmonic. "Junction parameters" as Fig. 2. $i_{dc} = 0.4$. $f_D = 210$ Hz.

bring out the odd subharmonics, thus the choice of 210 Hz in the present calculation ($\omega_D/\omega_p = 0.66$). On the analog all odd subharmonics up to $1/15$ have been observed. Naturally once a $1/n$ subharmonic is generated all higher harmonics of this will show up in the usual fashion. Returning to a $1/12$ subharmonic such may be produced in the following way. Consider a steady-state solution that includes a stable $1/3$ subharmonic. Insert this as v_0 in the Mathieu equation (Eq. (13)) and v_0 is seen to be possibly unstable towards a $1/6$ subharmonic. Thus one may generate a cascading bifurcation starting from an odd subharmonic as illustrated in Fig. 4 on the complete equation. So in a Josephson junction there is really an infinite choice of ways to bifurcate into chaos. To our knowledge these odd subharmonics have not been observed experimentally yet. However, this is probably because no one has ever arranged the excitation-detection system in such a way as to allow for the observation.

From the above discussion it is now clear why the $1/4$ subharmonic required the simultaneous presence of the $1/2$ subharmonic in the case of the Duffing equation. This is a reflection of the fact that the highest even order term in Eq. (19) is the second order term. For B_0 (or I_{dc}) $\neq 0$ a similar general statement as for the case of odd subharmonics applies for even subharmonics. Thus in principle a $1/vn$ subharmonic may spontaneously occur without the previous occurrence of the $2/n$ subharmonic. This, of course, will be generated as a harmonic.

Up to this point we have only considered the case of a spontaneous symmetry breaking correlated with the appearance of a time-independent term in the trial function. However, as mentioned before, another type of spontaneous symmetry breaking exists in the Josephson junction. This is the spontaneous jumps to finite voltage steps which may take place even with no dc current applied. This corresponds to including a term proportional to t in the trial function (i.e., a constant term in ϕ for the Josephson junction). For the Duffing equation the nonlinear term will then generate terms in higher orders of t which will then diverge. The Stewart-McCumber model on the other hand has a periodic potential. The divergencies are therefore successively cancelled by the higher order terms as is known from the usual Bessel function expansion for the step structure. Subharmonics of the pump frequency can therefore arise from a higher har-

monic of the finite voltage Josephson oscillation phaselocking with the pump. On these substeps (even or odd), the existence of which are well documented, one can then again have excitation of subharmonic frequencies, again both even and odd. We have e.g. observed a $1/3$ subharmonic of the fundamental Josephson frequency from a $1/5$ substep, i.e., $1/15$ subharmonic of the pump.

V. AMPLIFIER NOISE

In all externally pumped parametric amplifiers based on the Josephson junction one encounters the strange phenomenon of the noise rise which at least for tunnel junctions seems to have the unfortunate consequence that gain and noise temperature are proportional. For a detailed review of this we should refer to Ref. 10. For the microbridge at least the range of parameters is such that it seems very unlikely that chaotic noise might occur. The RC time constant corresponds to a frequency of order 500 GHz. The pump frequency is therefore well below the lower cut-off frequency for chaotic behavior. The phase-instability noise of Ref. 11 seems therefore a good candidate for explaining the noise rise here although some difficult points remain.¹⁰ We should remark here that although the exact coefficient is not yet known, a gain-dependent noise temperature is also found in the case of the microbridge. In what follows we shall devote ourselves to the question of tunnel junctions where chaotic noise may be expected to occur.

Since for the 4-photon amplifier one is actually able to choose junction parameters such that to operate outside the regime of chaotic behavior, we shall concentrate on the 3-photon amplifier in the following. There is of course no escaping that the operating point has to be close to the very threshold of $1/2$ harmonic excitation and therefore may be comparatively close to a regime of chaotic behavior, connected with a cascading bifurcation. In the calculation described below parameters have deliberately been chosen so that this is indeed the case.

In the 3-photon process one applies a signal of frequency $\omega_s \sim \omega_p \cos \phi_0$ which is the plasma frequency at the dc bias current $i_{dc} = i_c \sin \phi_0$. The pump frequency is then $\omega_D \sim 2\omega_s$. Turning up the pump amplitude one can then observe the parametric interaction through observing the idler $\omega_i = \omega_D - \omega_s \sim \omega_s$. For the chaotic noise to be a noise source in the parametric amplification one would require the idler and the chaotic noise to be present simultaneously. This, however, does not happen. The idler grows until the half-harmonic occurs, then the amplitude falls drastically. The amplitude then wobbles up and down as the $1/4$ subharmonic and the higher subharmonics set in but never becomes large again and long before the chaotic state occurs the idler has completely disappeared. This too happens in the four-photon case. This is, however, not conclusive evidence since a change in idler amplitude just may reflect an impedance change and bears no relation to amplification before input and output circuits are established. This has been realized on the analog computer in the following way (Fig. 5). Since the circulator isolates the signal source this is just a current source i_{in} to which one then can ascribe a source resistance equal to the load resistance. Since we are only interested in

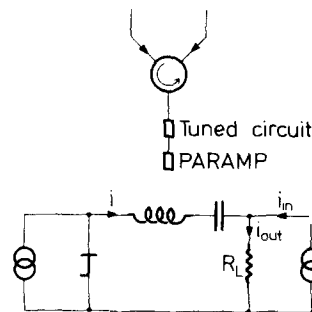


FIG. 5. Analog model of a circulator coupled Josephson junction parametric amplifier. The J stands for the Stewart-McCumber model of the amplifier.

the output signal at the input signal frequency, a tuned circuit can be used to dc isolate the load resistor from the junction circuit. Such a tuned circuit is actually often present in the experiments. The output i_{out} in the load resistor is now just the sum of the current in the tuned circuit and i_{in} . The amplification is then simply defined as $\Gamma = (i_{out}/i_{in})^2$. Proceeding as before gains above 30 dB were observed on the threshold of the half-harmonic. Small gain was obtained above threshold but as soon as higher order subharmonics were created, the gain disappeared completely and we in fact never obtained coexistent gain and chaotic noise.

With the idea in mind that external noise could switch the junction between states of amplification and chaotic noise and through this on the laboratory time scale make them look simultaneous, a white noise source (generated by the spectrum analyzer) was applied either directly to the junction or together with the signal. A typical example of the outcome is shown in Fig. 6. The double peak structure of the noise background is due to the tuned circuit¹² and is very often encountered in real life experiments.^{11,13} As can be seen we have amplified noise, but noise and signal are amplified together and the noise temperature is therefore constant; i.e., we have no noise rise. One should say that we do find a constant gain-bandwidth product given by $\Delta\omega/(\omega)\Gamma^{1/2} \sim 1/Q$ as expected from varactor theory.

For higher pump power values the half-harmonic is excited and the gain disappears. The output noise at the signal frequency for the curve with highest amplification is ap-

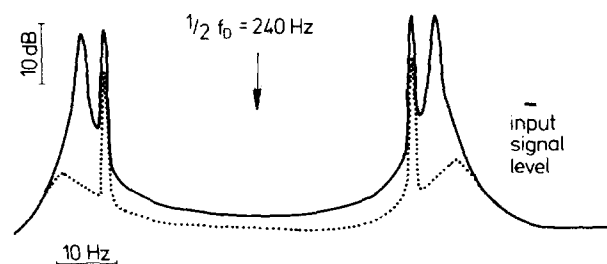


FIG. 6. "Output" of the amplifier model for two different "pump" amplitude settings. Lower curve $i_D = 0.230$, upper curve $i_D = 0.245$. Value of load resistor $4 \times R_J$, Q of tuned circuit (including load resistor) equals 10. "Junction parameters" as Fig. 2. $i_{dc} = 0.4$, $f_D = 480$ Hz, $f_s = 265$ Hz.

proximately equal to the chaotic noise level for yet higher pump values. But changing the input noise power makes no change in the qualitative picture. We should stress here that the actual shape of the noise spectrum far from the plasma frequency does not make any noticeable difference. Thus a high level of low frequency noise does not have much influence.

From this we conclude that the chaotic noise, although itself an intriguing phenomenon, does not seem to explain the observed noise rise neither for the 3- nor for the 4-photon amplifier. We have altogether not been able to produce the noise rise on the analog. One reason may be the comparatively low Q (~ 10) of the tuned circuit as compared to that in real experiments, where amplifier bandwidths of order 20 MHz are encountered compared to signal frequencies of 35 GHz. Such high Q values, however, are very difficult to realize on the analog. Also their occurrence in experiments is not fully accounted for. It is anyway difficult to see that a high Q could bring about the coexistence of gain and chaotic noise, but this could be the explanation for the failure of the mechanism suggested in Ref. 14 to show up.

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- ¹W. C. Stewart, Appl. Phys. Lett. **12**, 227 (1968); D. E. McCumber, J. Appl. Phys. **39**, 3113 (1968).
- ²B. A. Huberman, J. P. Crutchfield, and N. H. Packard, Appl. Phys. Lett. **37**, 750 (1980).
- ³R. L. Kautz, J. Appl. Phys. **52**, 3528, 6241 (1981).
- ⁴N. F. Pedersen, M. R. Samuelson, and K. Saermark, J. Appl. Phys. **44**, 5120 (1973).
- ⁵M. J. Feigenbaum, J. Stat. Phys. **19**, 25 (1978); J. Stat. Phys. **21**, 669 (1979); Los Alamos Science, Summer 1980, p. 4.
- ⁶M. J. Feldman, J. Appl. Phys. **48**, 1301 (1977).
- ⁷C. Hayashi, *Nonlinear Oscillations in Physical Systems* (McGraw-Hill, New York, 1964).
- ⁸M. T. Levinsen, R. Y. Chiao, M. J. Feldman, and B. A. Tucker, Appl. Phys. Lett. **31**, 776 (1977).
- ⁹M. T. Levinsen, N. F. Pedersen, O. H. Sorensen, B. Dueholm, and J. Mygind, IEEE Trans. Electron Devices **ED-27**, 1928 (1980).
- ¹⁰M. J. Feldman and M. T. Levinsen, IEEE Trans. Magn. **MAG-17**, 834 (1981).
- ¹¹R. Y. Chiao, M. J. Feldman, D. W. Peterson, B. A. Tucker, and M. T. Levinsen, in "Future Trends in Superconducting Electronics," edited by B. S. Deaver, Jr., C. M. Falco, J. H. Harris, and S. A. Wolf, AIP Conf. Proc. No. **44** (AIP, New York, 1978), p. 259.
- ¹²W. J. Getsinger and G. L. Matthaai, IEEE Trans. MTT-12, 77 (1964).
- ¹³Y. Taur, and P. L. Richards, J. Appl. Phys. **48**, 1321 (1977). B. Dueholm, N. F. Pedersen, O. H. Sorensen, J. Mygind, and M. T. Levinsen, Internal Report No. 206, DtH Lyngby.
- ¹⁴M. J. Feldman, and M. T. Levinsen, Appl. Phys. Lett. **36**, 854 (1980).